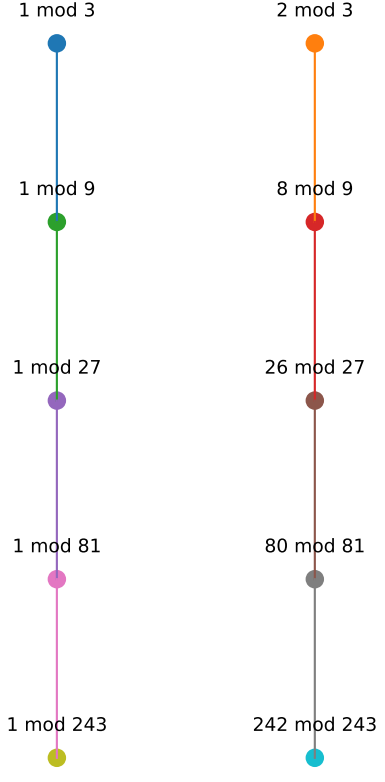
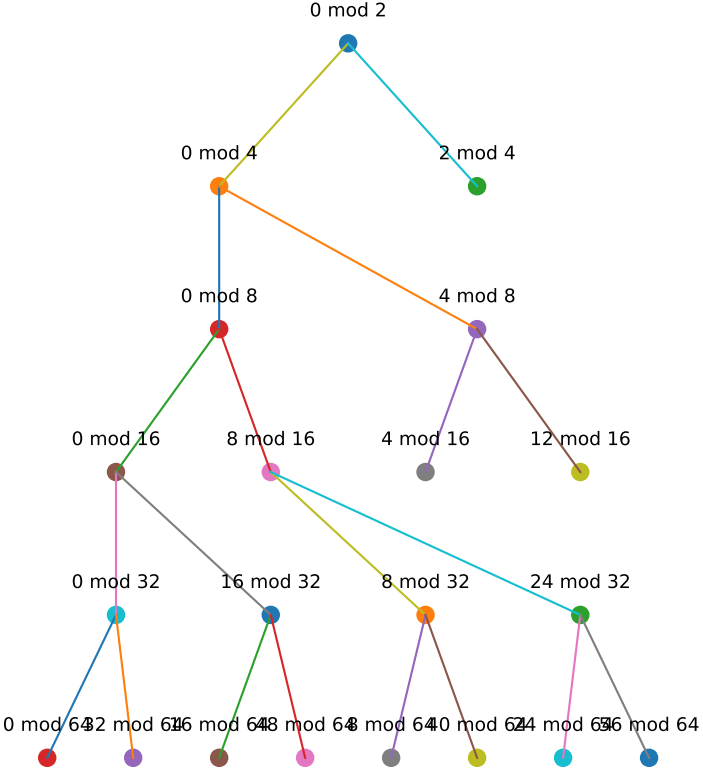
Congruence lifting exposes state-dependent branch geometry

$x^2 = 1$ over powers of 3
simple roots lift uniquely



$x^2 = 0$ over powers of 2
singular roots branch and die



The direction fiber at one residue is the finite set of valid next digits. Its size is an object property.